



**VietNam National University
University of Engineering and Technology**

EMBEDDED SYSTEM FUNDAMENTALS

(ELT3240, NHẬP MÔN HỆ THỐNG NHÚNG)

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Objectives

In this lecture you will be introduced to:

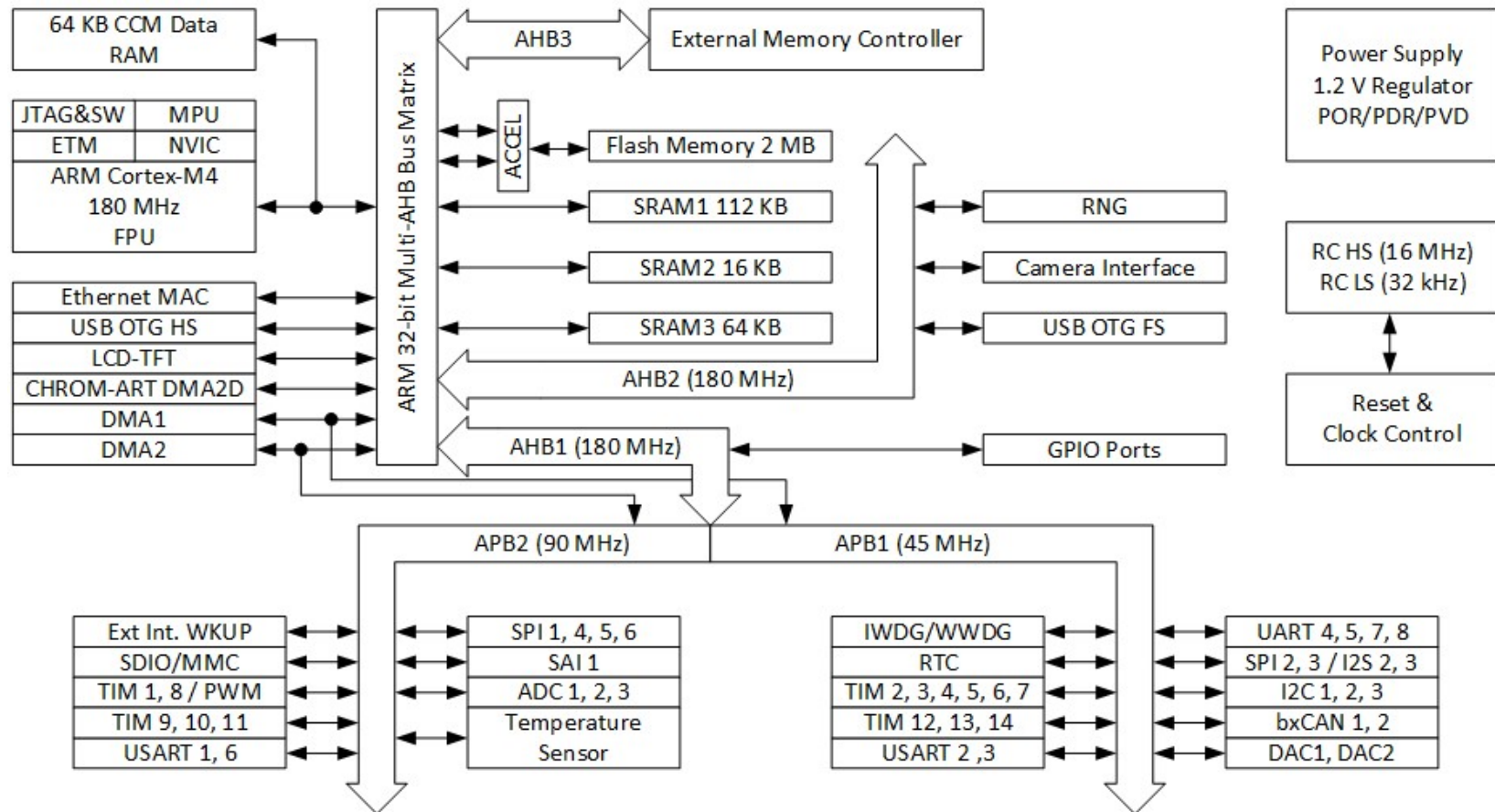
- A brief overview of the STM32F4 microcontroller architecture

Outline



- **The STM32F4 Microcontroller**
 - Central Processing Unit
 - Memory
 - General Purpose Input and Output Ports
 - Clock and Timer Modules
 - Analog Modules
 - Digital Communication Modules
 - Other Modules
- **The STM32F4 Board: Nucleo-F401re**
 - General Information
 - Pin Layout
 - Powering the Board and Programming the Microcontroller on It

The STM32F4 Microcontroller

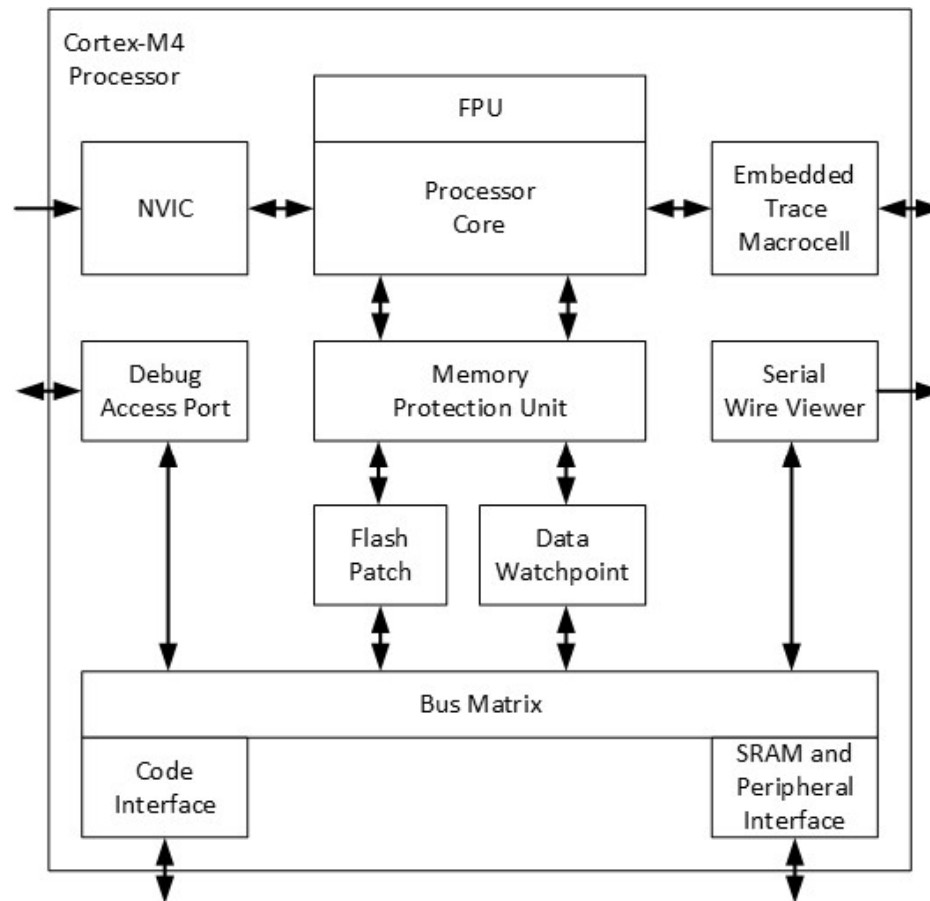


Functional Block Diagram of the STM32F4 Microcontroller

Central Processing Unit

- ❑ CPU is responsible for organizing all operations within the microcontroller.
- ❑ The programmer constructs his or her algorithm and forms the corresponding C, C++, or assembly code.
- ❑ This code is debugged and embedded into flash memory of the microcontroller by the help of an integrated development environment.
- ❑ The CPU executes the commands by using its resources (such as peripheral units).

Central Processing Unit



Block diagram of the CPU for the STM32F4 microcontroller

Central Processing Unit

CPU components

- ☐ Nested Vectored Interrupt Controller
- ☐ Debug Access Port and Serial Wire Viewer
- ☐ Memory Management Units
- ☐ Embedded Trace Macrocell Module
- ☐ The Bus Matrix
- ☐ Registers

Memory (memory)

Data to be processed and code to be executed is stored in memory of the microcontroller.

Memory Types

- Random Access Memory (RAM)
 - Stores temporary data.
 - Contents of RAM are lost when the power fed to it is turned off.
- Read Only Memory (ROM), Flash memory
 - stores code or data permanently.
 - Even if the power fed to it is turned off, contents of ROM are kept.

Memory

Vendor-specific memory	511 MB	0xFFFFFFFF
Private peripheral bus	1.0 MB	0xE0100000 0xE00FFFFFF 0xE0000000 0xDFFFFFFF
External Device	1.0 GB	0xA0000000 0x9FFFFFFF
External RAM	1.0 GB	0x60000000 0x5FFFFFFF
Peripheral	0.5 GB	0x40000000 0x3FFFFFFF
SRAM	0.5 GB	0x20000000 0x1FFFFFFF
Code	0.5 GB	0x00000000

The STM32F4 microcontroller
memory map

Memory

Memory Related Modules

- Direct Memory Access (DMA) module.
- Flexible Memory controller (FMC).
- Chrom-art accelerator controller (DMA2D) module.

General Purpose Input and Output Ports

- The microcontroller needs a medium to transfer data to and from the outside world.
- This is done by using ports of the microcontroller. A port is a collection of pins grouped together.
- Each pin can be taken as a wire with its electronic control circuitry.
- The STM32F4 microcontroller has eight input and output ports called A, B, C, D, E, F, G, and H. Total number of pins in these ports is 114.

Clock and Timer Modules

- With each clock cycle (MCLK), the processor performs an action that corresponds to an instruction phase.
- Cycles Per Instruction (CPI) is the average number of clock cycles required for a processor to execute an instruction.
- Millions of instructions per second (MIPS) is a unit used to characterize a processor's performance and corresponds to the processor frequency (MCLK) divided by the average cycles per second (CPI).

Analog Modules

- When we want to process an analog signal via the STM32F4 microcontroller, we will need an analog to digital converter (ADC).
- In the same manner, we will need a digital to analog converter (DAC) if we want to feed an analog circuit via the STM32F4 microcontroller.
- The STM32F4 microcontroller has specific modules to handle the ADC and DAC operations.

Digital Communication Modules

- Digital communication modules are used to communicate with external devices using dedicated communication protocols.
- The STM32F4 microcontroller has the following modules for digital communication.
 - three inter integrated circuit (I2C) modules
 - four universal synchronous/asynchronous receiver/transmitter (USART) modules
 - four universal asynchronous receiver/transmitter (UART) modules
 - six serial peripheral interface (SPI) modules
 - two controller area network (CAN) modules
 - one universal serial bus on-the-go full-speed (USB OTG_FS) module
 - one universal serial bus on-the-go high-speed (USB OTG_HS) module
 - one ethernet media access control (MAC) module with DMA

Other Modules

- There are also other modules under the STM32F4 microcontroller.
- We will not deal with them here.
- Please see the microcontroller user's guide for these.

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INTRODUCTION TO NUCLEO – F401RE

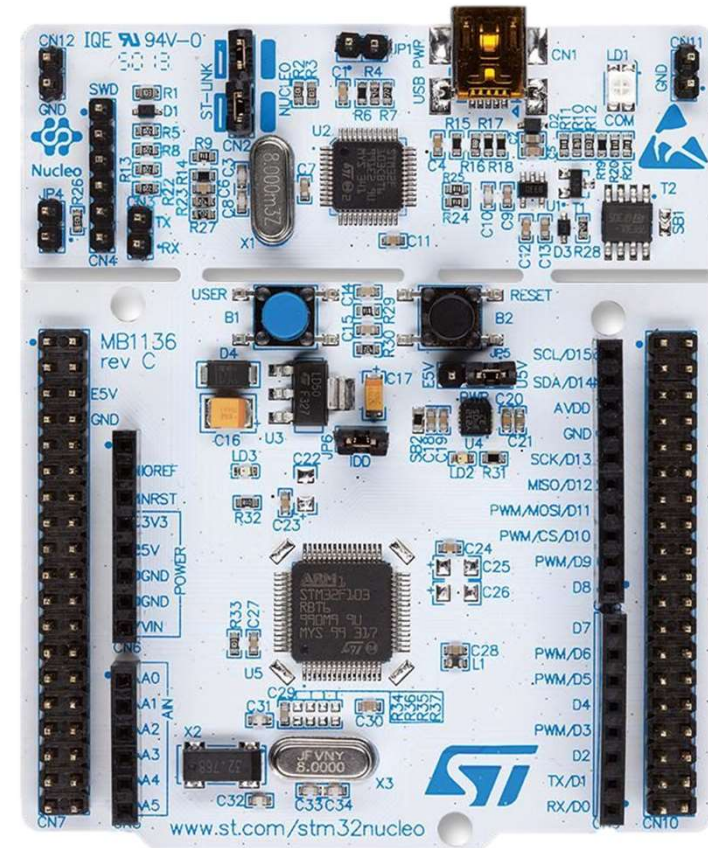
❖ BOARD FEATURES

Provides an affordable and flexible way for users to try out new concepts and build prototypes

STM32 microcontroller

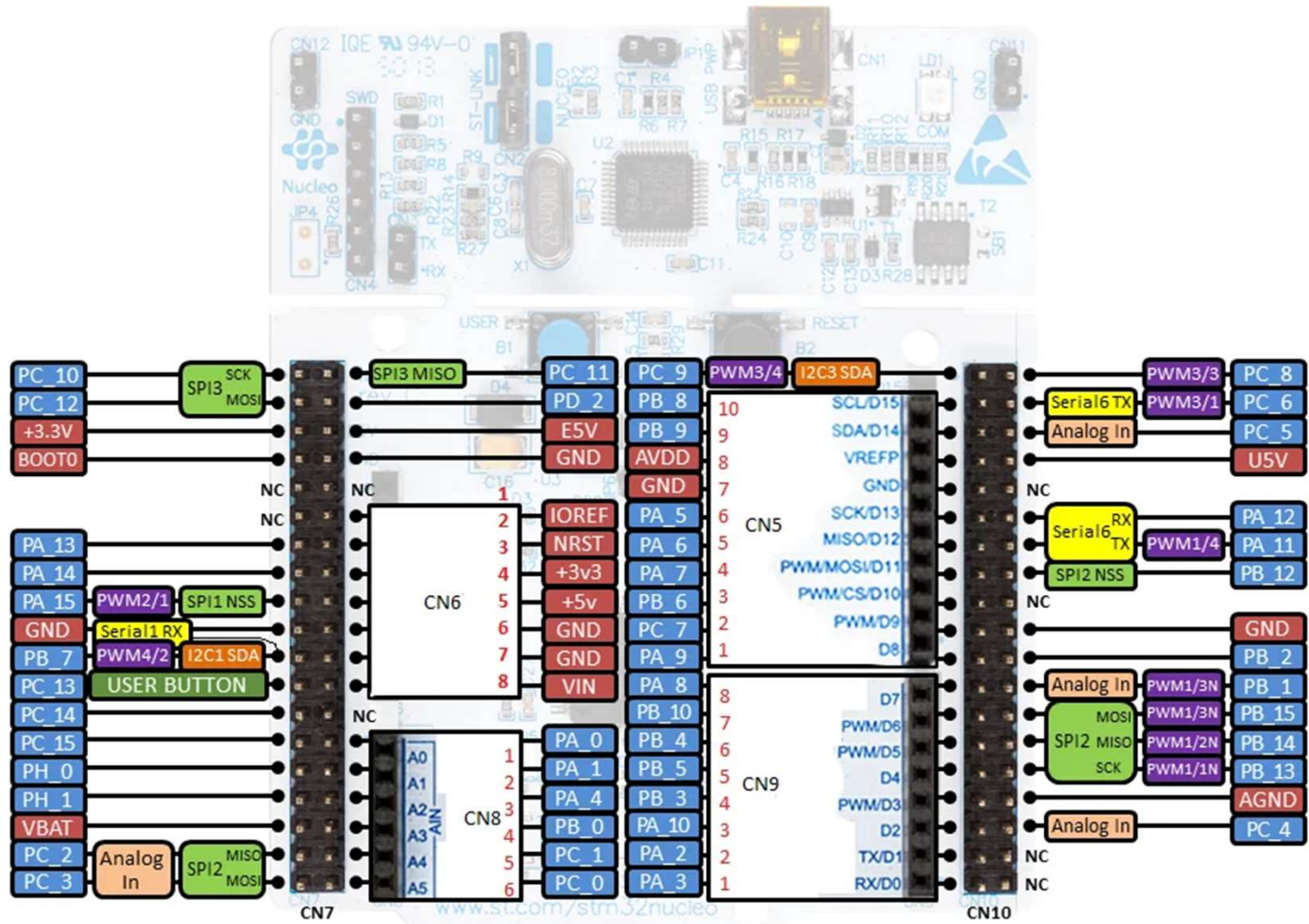
32.768 kHz crystal oscillator

- Two types of extension resources
 - Arduino Uno Revision 3 connectivity
 - STMicroelectronics Morpho extension pin headers for full access to all STM32 I/Os
- On-board ST-LINK/V2-1 debugger/programmer with SWD connector
 - Selection-mode switch to use the kit as a standalone ST-LINK/V2-1
- Flexible board power supply
 - USB VBUS or external source (3.3 V, 5 V, 7 - 12 V)
 - Power management access point
- User LED (LD2)
- Two push buttons: USER and RESET
- USB re-enumeration capability: three different interfaces supported on USB
 - Virtual Com port
 - Mass storage (USB Disk drive) for drag'n'drop programming
 - Debug port
- Comprehensive free software libraries and examples available with the STM32Cube MCU Package
- Support of a wide choice of Integrated Development Environments (IDEs) including IAR Embedded Workbench®, MDK-ARM, and STM32CubeIDE



INTRODUCTION TO NUCLEO – F401RE

❖ Pinout



The STM32F4 Board

Powering the Board and Programming the Microcontroller on It

- ❑ The microcontroller on the STM32F4 board can be programmed easily by the onboard ST-LINK/V2-B debugger/programmer. To do so, we should connect the board to PC via mini USB cable through its USB connector CN1.
- ❑ The USB connection for debugging/programming purposes can also be used to power the board.
- ❑ We can also use an external 5 V power supply to power the board.
- ❑ The STM32F4 microcontroller operates within voltage levels 1.8 V to 3.6 V.